

List

```
In [10]: l = (10,20,30,40,50,10)
l
```

```
Out[10]: (10, 20, 30, 40, 50, 10)
```

```
In [11]: l1 = (10,2.3,True,1+2j,'welcome',[1,2,3])
l1
```

```
Out[11]: (10, 2.3, True, (1+2j), 'welcome', [1, 2, 3])
```

```
In [12]: print(l)
print(l1)
```

```
(10, 20, 30, 40, 50, 10)
(10, 2.3, True, (1+2j), 'welcome', [1, 2, 3])
```

```
In [13]: l2 = [10,20,30]
```

```
In [14]: l2[0]
```

```
Out[14]: 10
```

```
In [15]: l2[0] = 100
l2
```

```
Out[15]: [100, 20, 30]
```

```
In [16]: l2.insert(2,25)
```

```
In [17]: l2
```

```
Out[17]: [100, 20, 25, 30]
```

```
In [18]: l2.insert(2,[1,2,3])
l2
```

```
Out[18]: [100, 20, [1, 2, 3], 25, 30]
```

```
In [19]: l2.pop()
l2
```

```
Out[19]: [100, 20, [1, 2, 3], 25]
```

```
In [20]: l2
```

```
Out[20]: [100, 20, [1, 2, 3], 25]
```

```
In [21]: l
```

```
Out[21]: (10, 20, 30, 40, 50, 10)
```

```
In [22]: l2.pop()
```

Out[22]: 25

In [23]: 1

Out[23]: (10, 20, 30, 40, 50, 10)

In [24]: 1

Out[24]: (10, 20, 30, 40, 50, 10)

In [25]: 12.pop(-2)

Out[25]: 20

In [26]: 11

Out[26]: (10, 2.3, True, (1+2j), 'welcome', [1, 2, 3])

In [27]: 1

Out[27]: (10, 20, 30, 40, 50, 10)

In [28]: print(l)
print(l1)
print(l2)

(10, 20, 30, 40, 50, 10)
(10, 2.3, True, (1+2j), 'welcome', [1, 2, 3])
[100, [1, 2, 3]]

In [29]: 11

Out[29]: (10, 2.3, True, (1+2j), 'welcome', [1, 2, 3])

In [30]: 1

Out[30]: (10, 20, 30, 40, 50, 10)

In [31]: 1.remove(10)

```
-----  
AttributeError                                Traceback (most recent call last)  
Cell In[31], line 1  
----> 1 1.remove(10)  
  
AttributeError: 'tuple' object has no attribute 'remove'
```

In []: 1

In []: 1.reverse()

In []: 1[::-1]

In []: 13 = ['a','b','c','d']
13

```
In [ ]: l3.reverse()
```

```
In [ ]: l3
```

```
In [ ]: l
```

```
In [ ]: l.sort()  
l
```

```
In [ ]: l5 = [3,200,23,100,10]
```

```
In [ ]: l5
```

```
In [ ]: l5.sort()
```

```
In [ ]: l5
```

```
In [ ]: l5.sort(reverse = True)
```

```
In [ ]: l5
```

```
In [ ]: l1
```

```
In [ ]: l.sort()
```

```
In [ ]: l
```

```
In [ ]: 10 in l
```

```
In [ ]: 100 in l
```

```
In [ ]: for i in l2  
print(i)
```

```
In [ ]: for l in enumerate:  
print(i)
```

```
In [ ]: any(l)
```

```
In [ ]: all(l)
```

```
In [ ]: l2.append(0)
```

```
In [ ]: l
```

```
In [ ]: any(l)
```

Tuple

```
In [ ]: t = ()  
t
```

```
In [ ]: type(t)
```

```
In [ ]: t1 = (10, True, 2.3, 1+2j, 'welcome')  
t1
```

```
In [ ]: t1.append(50)
```

```
In [35]: t1 = (10, True, 2.3, 1+2j, 'welcome')  
t1
```

```
Out[35]: (10, True, 2.3, (1+2j), 'welcome')
```

```
In [36]: t1[0]
```

```
Out[36]: 10
```

```
In [37]: t1[0:4]
```

```
Out[37]: (10, True, 2.3, (1+2j))
```

```
In [38]: t1[0]
```

```
Out[38]: 10
```

```
In [39]: t1[0] = 100
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[39], line 1  
----> 1 t1[0] = 100  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [40]: t1.count(10)
```

```
Out[40]: 1
```

```
In [41]: t1.index(1+2j)
```

```
Out[41]: 3
```

```
In [42]: t1
```

```
Out[42]: (10, True, 2.3, (1+2j), 'welcome')
```

```
In [43]: t1.remove(10)
```

```
-----  
AttributeError                            Traceback (most recent call last)  
Cell In[43], line 1  
----> 1 t1.remove(10)  
  
AttributeError: 'tuple' object has no attribute 'remove'
```

```
In [44]: t = (10, 20, 30)  
t
```

Out[44]: (10, 20, 30)

In [45]: `t * 3`

Out[45]: (10, 20, 30, 10, 20, 30, 10, 20, 30)

In [46]: `sbi = (1234, 'abcdefgh', 20000, 'Bangalore')`
`sbi`

Out[46]: (1234, 'abcdefgh', 20000, 'Bangalore')

In []: